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By electronic transmission

Mr. Robin Li
Chairman and Chief Executive Officer
Baidu, Inc. — Apollo Go
Beijing, People's Republic of China

Re: Exploratory inquiry — fleet-level fail-safety and independently checkable conformance for Apollo Go's international expansion

Dear Mr. Li:

I am the founder of Ghost Autonomy, a pre-silicon research program based in Halifax, Canada, developing what we call an **enforcement-and-accountability layer for bounded autonomy**: a risk-monotone actuation boundary terminating in non-programmable analog hardware, an evidence-admissibility discipline applied *before* sensor fusion, and a constitutionally governed memory in which every change to a system's operating truth is a logged, challengeable event. The program's distinguishing practice is evidentiary: every claim in our corpus carries a graded standing — measured, proven, proposed, or notional — and no claim is permitted to exceed it. This letter observes the same rule.

Apollo Go operates the largest robotaxi program in the world by cumulative rides — more than 22 million across 27 cities, on a purpose-built vehicle whose sub-\$30,000 cost defines the industry's economic frontier — and in June 2026 it earned a Level-4 permit from the Swiss federal roads authority, among the first of its kind in Europe. We write because scale of this kind changes what safety must mean. The Wuhan event of March 31, 2026, in which vehicles halted simultaneously on a platform-level fault, demonstrated something no operator had shown at scale before: in a coordinated fleet, the unit of failure is the platform, not the vehicle — and the industry's minimal-risk architecture is still defined per vehicle. At the same time, your European and Middle-Eastern expansion brings Apollo Go before assessors and regulators whose regimes are built on independently checkable safety cases rather than aggregate operational statistics. In both directions, the set of assumptions on which the program continues is now the critical object.

At this point in your program's trajectory, we believe the most consequential engineering object you own is the set of assumptions on which it will continue — and that this set now deserves explicit registration, physical enforcement, and independent audit.

The offer. Ghost is prepared to pivot its development to maximize compatibility with the Apollo Go stack: our conformance instrument formatted to the Swiss and UNECE assessment grammars your European deployments will answer to; our guard-ECU embodiment engineered against a bill-of-materials ceiling consistent with the RT6 cost curve, not a data-center accelerator; and our witness-log design aligned natively with the data-recorder requirements China's national rules make mandatory from July 2027.

What Ghost brings — specifically, and at its true standing.

1. **A measured boundary primitive.** An electrically isolated analog safety veto with ~ 32 ns intervention latency, witnessed on FPGA **[MEASURED]** — deliberately the only latency figure our corpus reports as measured rather than projected.
2. **Proven envelope mathematics.** The risk-monotone actuation algebra in its box case: projection is coordinatewise clipping; enforcement is firmly non-expansive (1-Lipschitz); the two-stage clamp is exact in the feasible case **[PROVEN]**.
3. **A fully specified refusal chain.** Five stages (S0–S4), from trusted numerics through fidelity gating, layered safety filtering, and risk-monotone envelopes to the analog veto, with six acceptance criteria (G1–G6) restated as a conjunctive, fail-closed conformance gate — published as criteria, not achievements **[PROPOSED]**.
4. **An admissibility-before-fusion evidence discipline.** Sensors as witnesses under integrity, alignment, scope, and lineage gates; correlated witnesses counted once; inadmissible evidence excluded rather than averaged in — sensor-agnostic by construction **[PROPOSED]**.
5. **A constitutional memory doctrine.** Append-only canon, anti-silent-drift amendment semantics, verification burden that increases with residence time, and recall as a priced contract (Governed Knowledge Systems trilogy, July 2026) **[NOTIONAL]**.
6. **A self-auditing corpus.** An 89-page consolidated foundational corpus (May 2026); a 94-page technical documentation edition, v3.0 (June 2026); and a 367-page compendium whose provenance ledgers grade every claim — including a published coverage map (68 subcategories: 21 full, 28 high, 9 partial, 10 gap) and a formal proof-debt ledger (P-1–P-7) naming what remains unproven.
7. **Empirical provenance, honestly scoped.** Physics-informed reduced-order kernels on elliptic-integral/AGM primitives with machine-precision convergence **[PROVEN]**, validated at $R^2 \approx 0.78\text{--}0.87$ on 200+ experimental datasets within their demonstrated envelope **[MEASURED]** — in a non-automotive domain, with automotive transfer registered in our own ledger as an open validation obligation.

What this could be worth to your program — specifically.

1. **Fleet-level minimal-risk semantics.** Correlated-witness discounting and constitutional quarantine of suspect platform updates — an architecture addressed to the failure mode of March 31, which per-vehicle redundancy does not reach.
2. **Independently checkable conformance verdicts.** Signed witness logs and replayable decision traces allow a European assessor to verify claims without trusting the operator — converting company-reported safety into evidence, the currency in which your Swiss and EU deployments will be priced.
3. **A hardware refusal boundary at component economics.** Designed from the outset for a sub-\$30,000 vehicle's cost structure, so that the fleets that scale fastest are the fleets that can afford it.
4. **Native alignment with the 2027 data-recorder mandate.** The append-only decision canon is, by construction, the black box the rules require — with audit and challenge semantics beyond them.

What Ghost is not. We have no fabricated silicon, no vehicle deployment, and no fleet data; our measured hardware evidence is presently a single number. We state this plainly because we regard the statement itself as an asset — a partner adopting our layer inherits a provenance

ledger, not a marketing narrative — and because it is precisely why a compatibility-first alignment now, while our interfaces are still being fixed, costs little and shapes much.

Our inquiry is deliberately modest: *is such an alignment of interest?* As a first, zero-obligation step, we would run our conformance instrument against a trace or scenario format of your choosing and return the results with their standings — an artifact your team keeps regardless of what follows. Our two strategy volumes and the published corpus are available on request; deeper documentation under mutual NDA.

Respectfully,

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Evidentiary standings used in this letter — **[MEASURED]** witnessed on hardware; **[PROVEN]** a mathematical result; **[PROPOSED]** specified and defended, not yet validated; **[NOTIONAL]** a research direction, quotable as intent only. It is our house rule that no statement may exceed its standing; we extend the same rule to this letter.